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Design of Artificial Intelligence-Based Products: Barriers and Enablers

Abstract

Artificial Intelligence (AI) embodied products are becoming ubiquitous in the modern world. Organisations are hence updating themselves to design and develop such products. In this paper, we aim at identifying enablers and barriers in designing such products across sectors. Our analysis of a broad range of literature in this field allowed us to identify these enablers and barriers. We have developed SOTCUT and SEECUT models representing these factors. We have discussed implication of the findings for the practice of designing AI-embodied products that could be useful to the related industry.

Keywords: Artificial intelligence (AI), Design practice, Barriers, Enablers

1. Introduction

Artificial Intelligence (AI) is a rapidly growing field and has made significant progress in the last few years. Research in AI can be traced back at least to the 1950s (Russell and Norvig, 2021). Recent developments in the field have paved the way for the successful integration of AI in designing AI-embodied solutions and transforming several domains. In 1997, IBM's Deep Blue, an AI-based program, defeated the chess champion Garry Kasparov. More recently, in 2017, Sophia – a robot using AI – was awarded citizenship by Saudi Arabia. Whilst there is absence of a commonly accepted definition of AI, it is loosely defined as a "system's ability to correctly interpret external data, to learn from such data, and to use those learnings to achieve specific goals and tasks through flexible adaptation" (Kaplan and Haenlein, 2019). Nonetheless, it is now widely accepted that AI systems have revolutionized current technology. AI is supporting humans to perform several tasks effectively and efficiently and even executing some tasks without any need for human intervention.

The applications of AI are varied, ranging from energy (Jha et al., 2017), healthcare (Dreyer and Allen, 2018; Kahn, 2017), transportation (Abduljabbar et al., 2019), construction (Muhuri, Shukla, & Abraham, 2019; Parveen, 2018), manufacturing (Li et al., 2017; Haeffner and Panuwatwanich, 2017), design (Wei, 2018), education (Mikhaylov et al., 2018) etc. In fact, in the modern context there is virtually no field that is untouched by the presence of AI. Design is essential to

develop solutions that use new technologies, including AI (e.g. Cross, 2021; Eppinger and Ulrich, 2015).

While AI has been making rapid developments in the last decade, researchers have only recently started to focus on more rigorous studies of its impact and evaluation on the society and vice-versa. However, the work investigating the design of AI-embodied products has largely remained fragmented. AI-based solutions are present in varied fields, with each field having a set of factors that support or hinder design and development of such solutions. Knowledge on these factors across several fields helps to develop suitable tools and methods for supporting organizations and designers of AI-embodied solutions. However, there is a lack of knowledge on these factors. This limits our ability to develop ways to alleviate hindering factors or to leverage on supporting factors.

In this paper, we have identified and developed a model of factors that support or hinder design, development, and implementation of AI-embodied products in various fields. These factors have been extracted based on the analysis of extant literature in the field. These factors have been framed as SOTCUT and SEECUT models, referring to the enablers and the barriers respectively. The enabling factors identified in SOTCUT model include issues pertaining to Success stories, Organizational support, Training programs, Collaboration, User requirements and Technology. Similarly, the barriers in SEECUT model refer to issues pertaining to Society, Economy, Ethics, Collaboration, User requirements and Technology.

The remainder of this paper is structured as follows. Section 2 lays out the findings of the analysis of the relevant literature in the field, providing a basis for the SOTCUT and SEECUT models. Section 3 discusses the findings. Section 4 presents conclusions, while highlighting limitation of this work and suggesting areas for future research.

2. Findings

We first identified studies that have investigated design, development, and implementation of AI-based solutions in a wide range of fields such as energy, transportation, healthcare, etc. Our focus was on understanding the barriers and enablers that influence the design and growth of AI in different domains. Our goal was to present a consolidated cross-domain framework of these supporting and hindering factors in shaping the AI-embodied solutions.

2.1 Enablers

Recent surge in AI has given insights into various aspects of production and consumption of this technology, that were hitherto unexplored. Our findings suggest

that a broad range of factors support design of AI-embodied solutions. These factors are related to the designers of AI-based solutions as well as to the consumers of these solutions. Furthermore, equally important are social and technological aspects.

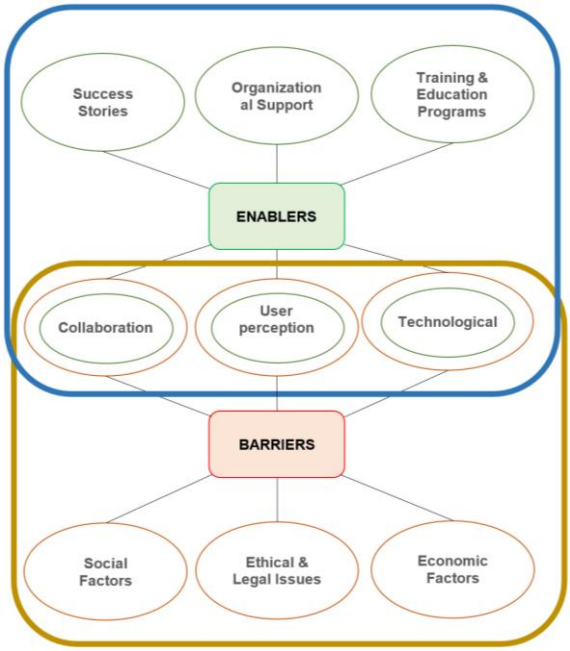


Fig. 1. SEECUT and SOTCUT framework

2.1.1 Success stories

Potential opportunities of AI and resulting solutions, their superior outcomes and performance can support their development and implementation (Dwivedi et al., 2021). Successful demonstrations of beneficial results of AI-based solutions have a positive impact on their design and development. For example, the modern world is planning to rely heavily on the development of autonomous vehicles based on the AI technology (Abduljabbar et al. 2019). Such vehicles are expected to contribute by enhancing productivity and reducing the number of accidents on highways. Autonomous trials of small-scale buses and self-driven cars in Finland, Singapore have shown promising results. In addition to the development of autonomous driving vehicles, the use of AI in corporate decision making, planning and management could be valuable, as discussed by Abduljabbar et al. (2019). For instance, AI-driven planning systems could help to plan resources better to overcome or partially mitigate

the issue of a continuously rising transportation demand with limited road supply. This field is broadly known as Intelligent Transport Systems (ITS).

With the advent of AI, an increasing emphasis is being given to its use in sustainability and judicious consumption of resources (Jha et al. 2017). This is gaining more importance in modern world with the rise of energy consumption and living standards. Furthermore, their findings reveal widespread application of AI in several fields of renewable energy generation or tapping, for example, wind, solar, geothermal, hydro, ocean energy etc. This is a key enabler that hints at possible scalable application of AI across several domains and fields.

2.1.2 Organizational support

Implementation of AI-based products is supported by appropriate al strategies (e.g., Strohm et al., 2020). Organizations having specific innovation strategies aimed at promoting AI have beneficial effects on realization and implementation of AI-based products. Presence of an innovation manager or a local champion promoting and facilitating the implementation of AI products has positive impact on adoption and sustained use of AI applications (e.g., Strohm et al., 2020). Local champions can play a crucial role in implementation of AI-based products within their organizations, while taking the lead in the full process covering various activities. Such local champions may have a keen interest in AI and its applications and may possess sufficient knowledge of various aspects of AI applications. They may need to overcome opposition from colleagues who may be skeptical about AI products and services. Proving information on AI and offering opportunities to experiment with AI applications can overcome fear and skepticism about AI (e.g., Strohm et al., 2020).

2.1.3 Training and education programs

Appropriate university programs and courses aimed at developing an aptitude among students to design, develop and implement AI enabled solutions have beneficial consequences on the development of future experts and practitioners in the field. Institutional or country-level policies and programs assisting the inclusion of AI in curriculum also supports the development of future professionals skilled in developing and implementing products and services that embody AI to satisfy some functions (e.g., Strohm et al., 2020). For example, Strohm et al.'s (2020) study found that a platform to exchange knowledge in a specific area of healthcare sector can support implementation of AI technologies, can provide guidance to hospitals in developing and using AI strategies, and can promote inclusion of AI in the curriculum. Recht et al. (2020) suggest that collaborative development of AI educational programs and curriculum requires joint efforts of a wide range of stakeholders such as national and international committees and academic departments.

In order to develop AI curriculum and educational resources in a specific domain, it is essential to identify the skills and knowledge requirements of future practitioners who will implement and use AI-based solutions in that domain. For example, if future practitioners in the field of radiology need to use AI-based solutions in their daily practice, they may need to possess basic proficiency in machine learning, while having awareness of advantages and limitations of related AI solutions (e.g., Wood et al., 2019; Tajmir and Alkasab, 2018).

2.1.4 Collaborative design

Collaboration between a wide range of stakeholders can support various activities in the process of designing and implementing AI-based solutions. For example, designers and potential users of such tools can work with ethicists to support ethical use of AI-based tools (e.g., Recht et al., 2020). As such, effective collaborative design is considered as an enabler in the process of designing, developing, and implementing AI-based solutions.

2.1.5 User requirements and perceptions

AI-based solutions can create impact if they are accepted and used (e.g., Edwards et al., 2000). Identifying and implementing requirements about ergonomics and interaction of humans with AI can positively influence acceptance and use of AI-based solutions. Creating and disseminating appropriate narratives about AI-based solutions have a beneficial impact on their adoption by the intended users. For example, reframing professional roles and identity of radiologists as "imaging consultants" can positively influence the acceptance and use of AI enabled solutions in the field of radiology (Strohm et al., 2020). Formulating appropriate and context-specific narratives can thus support changes in perception about roles of target users.

To enhance acceptance and adoption of AI enabled products, they need to be user-friendly. Such products are perceived as user-friendly when they are designed by considering existing systems in which they will be integrated and used (e.g., Strohm et al., 2020). The design of these products ought to consider requirements of their easy integration in existing systems, for example, by eliminating additional changes to existing systems.

2.1.6 Technological

Amount of data: In recent years, collection, and storage of large amount of data has become feasible due to the availability of various technologies such as the Internet of Things (IoT) and collection of data logs. This data can be analysed to

extract patterns which can be used for many other purposes. The need of analysing large data to reveal hidden patterns has a beneficial impact on the design of AI-based products that can analysed large data to glean important insights. For example, in some domains, data logs sent to the server contain valuable information on the desirable and missing features of the products and services in those domains (Nishant et al. 2020). Such massive, unstructured data logs are the source of big data, which is processed with the help of machine learning or data mining. Dealing, managing and analysing this magnitude of data is inconceivable without the design and development of related AI-based solutions.

Access to diverse datasets: Availability of diverse datasets also supports development of AI-based solutions. For example, in the case of healthcare sector, datasets need to be representative, requiring various institutions to share data from various demographics and countries. Access to such large and diverse datasets not only supports the design of AI-based solutions but also aids training and evaluation of these solutions by using diverse datasets, enhancing generalisability of AI-based solutions (e.g., Recht et al., 2020).

2.2 Barriers

Since AI is a more recent technology that has infiltrated in various sectors, it is likely to encounter a set of hidden facets that could be potential barriers. Several factors play a hindering role when considering incorporation of AI in different fields. In addition to the apparent economic issues, several other social, technological, and ethical issues gain prominence in this process.

2.2.1 Social factors

Crawford et al. 2016 consider the possibility that even though AI-based systems are designed to empower workers, there are substantial ways in which these technologies could end up disempowering workers and even generate unfair labor practices.

Employment: AI enabled solutions can have profound impact on employment and roles of humans. Therefore, these issues as well as requirements about re-skilling employees need to be considered in developing AI-based solutions. Using probabilistic analysis, the World Economic Forum (WEF) estimate that about 20% of existing jobs in the UK could be affected by AI technologies, and that this percentage is greater in countries such as China and India (Dwivedi et al 2021). In addition, in emerging markets, the lack of education can hinder implementation and adoption of AI enabled solutions. The governments in these countries will need to design solutions and strategies for embracing AI technologies, while managing the impact of these solutions on employment.

Transparency and accountability: Transparency and accountability in the decisions made by AI-based solutions is a challenge in developing such solutions. There are various challenges in offering explanation behind a specific decision made by such solutions (Miller, 2019 in Dwivedi 2021). For example, it is difficult to estimate accuracy of reasoning offered to explain a specific decision. Furthermore, biased, and non-transparent AI algorithms can lead to negative consequences on expectations of transparency and liability (e.g., Dwivedi et al., 2021).

Data protection and availability: The design and use of AI solutions in some sectors can be hindered by the concerns related to data protection and privacy. For instance, seeking consent, anonymization of data can be challenging when activities in the design and development of AI algorithms require access to large datasets representing people from a wide range of backgrounds (e.g., Balthazar et al., 2018 in Recht et al., 2020). Furthermore, lack of data or difficulties in gaining data representing diversity in a society can be challenging not only in validating and generalizing AI solutions, but can also deepen the inequalities in the society, for example, when the datasets from disadvantaged and vulnerable people are missing. Such inequalities can also be due to differences between implementation of AI solutions high-income and low-income societies (Schönberger, 2019 in Recht et al., 2020).

Marginalization: Crawford et al. 2016 investigates another barrier that has societal impacts when deploying AI technologies in various sectors and devices. On the user end, AI technology might favor the population of certain sections, income groups or ages. This could widen the already existing socio-economic gap in the society.

A crucial ingredient for successful urbanization is digitization (Allam & Dhunny 2019). However, older citizens often have a lag towards digitization (both in developed and developing worlds), which is perceived as a barrier to AI-based urban efficiency and performance.

2.2.2 Ethical and legal issues

Advances in AI are raising challenging ethical and legal issues. Governments play a key role in addressing such critical issues, and thereby in supporting design and development of novel AI-based solutions that can support societal development. Inadequate policies, guidance, and framework to address ethical concerns about AI solutions has a problematic affect not just on the development of such solutions, but also on their impact on societies.

Lack of regulatory frameworks or uncertainties in regulatory and legal procedures for AI-based solutions has a detrimental effect on realization and implementation of such solutions. For example, a study investigating implementation of AI enabled solutions in healthcare sector found that unaddressed questions regarding legal implications of damage that might occur due to products and services using AI has a limiting effect on their development and implementation (Strohm et al.,

2020). In addition, difference in legal procedures between different countries and regions can also pose a challenge in developing and implementing AI solutions (e.g., Recht et al., 2020).

The development and implementation of AI-based solutions need to consider public policy requirements (e.g., Dwivedi et al., 2021). The high speed of development in AI technologies provides insufficient time public policy practitioners to reflect and respond to their impact on various aspects.

2.2.3 Economic

The economic aspect while designing and developing AI-based approaches comes into play from multiple angles.

Funding: Lack of essential funding has a problematic affect not only on activities of designing AI-based solutions but also on their implementation and adoption by the intended recipients (Strohm et al., 2020). Sector and organization specific factors can also hinder development and implementation of AI enabled solutions due to lack of funding. For example, as compared to large organizations, SMEs can face challenges in implementing such solutions. Small organizations can face several challenges ranging from data quality, security risks to lack of experts in AI technologies (e.g., Dwivedi et al., 2021).

Developing cost: On the developer end, creation, training, and deployment of AI systems require a huge amount of data, power, infrastructure and other resources. This could exclude many actors (for example new entrants), who cannot afford the inclusion of these technologies in their products.

2.2.4 Collaboration

Nishant et al. 2020 points that in context of AI, in order to develop solutions to real-life problems, researchers and practitioners should work together. This is likely to bridge the existing gap between design thinking methods and in-field, design science research (Denning 2013). Abduljabbar et al. 2019 in their work, have given the example of ITS wherein a mechanical intelligence needs to be blended with the proper understanding of human-based information (Waltz 1997).

2.2.5 User issues

User perceptions: The ways in which the potential users of AI-based solutions view and perceive these solutions influence the acceptance and adoption of these solutions (e.g., Dos Santosh et al., 2019 in Strohm et al., 2020; Strohm et al 2020). For example, Strohm et al.'s (2020) study of AI-based solutions in the field of

clinical radiology found that the perspectives of radiologists might range from curiosity to fear.

Negative perception about AI technologies can also have a limiting effect on adoption and use of AI enabled products and solutions (e.g., Dwivedi et al., 2021). For example, for some people, the AI concept is difficult to understand, and they can find it difficult to understand how it can affect their everyday activities. Such perceptions can lead to confusion and frustration, creating challenges for how to design AI-based solutions for supporting intuitive and natural interaction between them and humans.

Far-fetched and unrealistic expectations of AI can potentially be harmful for the design and solutions that embody AI to fulfil some functions (e.g., Dwivedi et al., 2021). The pattern of rise and fall in hype and expectations can create ambiguity and confusion in the potential of AI, and thus may hinder the development of AI enabled solutions.

Training: In the case of some AI-based products, the users of such products may require some technical understanding to assess quality and safety of the products (e.g., Char et al., 2018; Rubin, 2019 – both in Strohm et al 2020). Lack of this technical understanding can hamper the implementation of AI-based products.

2.2.6 Technological

Generalizability: Difficulties in enhancing generalizability of AI solutions has a problematic effect on their widespread implementation and adoption in a diverse range of settings in a specific domain or across domains. Algorithms designed in an AI-based solution and trained on a particular dataset can yield superior performance on similar data but may exhibit poor performance when they use data on which they are not trained (e.g., Zech et al., 2018 in Recht et al 2020).

Poor prediction: A key factor to successful machine learning (ML) prediction models is the historical data on which it is trained. However, if the ML models are highly simplified or biased, it can lead to a poor performance both in case of training and test data (Blekin et al. 2019). Similarly, with the changing times, past data always does not always reflect the future trend, thereby possibly leading to poor predictions (Nishant et al. 2020).

Technical performance/Quality: Technical performance of AI enabled solutions influences their acceptance and use. Inconsistent technical performance has negative consequences on how users perceive and interpret these solutions (e.g., Strohm et al., 2020). For example, in the case of healthcare sector, many false positives or false negatives have detrimental effect on implementation and adoption of AI enabled healthcare technologies and products.

Lack of evidence on benefits of AI: Duan et al. 2019 identify the issue of lack of direct metric to measure the influence or benefit of AI in a field. The lack of evidence to support beneficial outcomes of AI solutions has a limiting effect not only on developing new AI enabled applications but also on implementation of

developed solutions. Furthermore, measuring various benefits of AI-based solutions is difficult. This problem is further exacerbated by the lack of standard methodologies to measure various consequences of products and services that use AI (Strohm et al., 2020).

Data sharing difficulties: Several aspects that can inhibit data sharing can have a problematic effect on testing, implementation, and adoption of AI-based solutions (e.g., Recht et al., 2020). For example, there is a relative lack of training programs for developers of AI solutions about the benefits and difficulties of data sharing. This problem of data sharing is further exacerbated by the significant resources that are required to install and maintain data sharing systems and platforms. In addition, competition between different academic institutes can have negative consequences on data sharing.

3. Discussion and Conclusions

A broad range of factors support the design, development, and implementation of AI-based products. While some factors relate to technological aspects, others relate to the ways in which AI-based products and services are designed and developed. For example, a large amount of data is generated by modern devices and systems. This data has created an important need to reveal underlying patterns in order to glean insights which can further be used in designing new products or in customizing existing products to serve needs of various type of customers. With the growing maturity of AI technology, identifying and implementing user requirements is likely to play a key role in designing and shaping the attributes of future AI products. In addition, successful demonstrations and results of AI-based products in various sectors such as energy, healthcare, transportation, etc. stimulates further development of AI enabled products and system in those sectors as well as other sectors. This includes exploration of AI solutions that are sustainable. In the last decade, there has been an ongoing transformation of educational and training programs in AI and related fields. Such programs are increasingly catering to the growing demand of AI experts and professionals in the industry and other s.

Just as several factors support the design and development of AI solutions, a broader range of factors acts as barriers. The cost of AI embodied product development, including access to data, can have a limiting effect on s targeting such products. This also has adverse impact on product affordability for customers. Furthermore, not only current AI solutions are ungeneralizable across different domains, but they are also in some cases negatively affected by the absence of a broad historical data for accurate prediction. The field experts are also confronted with the challenge of achieving a high performance of designed AI solutions so that they can be good fit for real world applications. While the AI field itself is developing rapidly, there is absence of standard qualitative or quantitative metrics to measure technical performance of the AI, especially from the user perspective. In a similar fashion,

social factors can have limiting effect on the design and adoption of AI embodied products. There has been a growing concern that the automation resulting from AI solutions can reduce employment opportunities. In addition, to encourage development and adoption of AI solutions, there is a need to address the concerns regarding data protection raised by societies and governments. Furthermore, lack of appropriate rules, legal frameworks, and specific ethical guidelines hinders the development and implementation of AI embodied solutions.

Whilst we scrutinized and analyzed a broad range of literature sources to develop this model, future research can gain by undertaking empirical work aimed at investigating and identifying additional (if any) barriers and enablers. Although our findings are generic, future studies can benefit from domain-specific studies to uncover barriers and enablers in a particular domain. In addition, further studies can compare factors affecting design of AI solutions across different domains.

References

Abduljabbar R, Dia H, Liyanage S, Bagloee SA. Applications of artificial intelligence in transport: An overview. Sustainability. 2019 Jan;11(1):189.

Abduljabbar, R., Dia, H., Liyanage, S., & Bagloee, S. A. (2019). Applications of artificial intelligence in transport: An overview. Sustainability, 11(1), 189.

Allam Z, Dhunny ZA. On big data, artificial intelligence and smart cities. Cities. 2019 Jun 1;89:80-91.

Balthazar P, Harri P, Prater A, Safdar NM (2018) Protecting your patients' interests in the era of big data, artificial intelligence, and predictive analytics. J Am Coll Radiol 15(3):580–586

Belkin M, Hsu D, Ma S, Mandal S. Reconciling modern machine-learning practice and the classical bias–variance trade-off. Proceedings of the National Academy of Sciences. 2019 Aug 6;116(32):15849-54.

Char DS, Shah NH, Magnus D (2018) Implementing machine learning in health care ' addressing ethical challenges. N Engl J Med 378(11):981–983.

Crawford K, Whittaker M, Elish MC, Barocas S, Plasek A, Ferryman K. The AI Now Report. The Social and Economic Implications of Artificial Intelligence Technologies in the Near-Term. 2016.

Cross, N. (2021). Engineering design methods: strategies for product design. John Wiley & Sons.

Denning PJ. Design thinking. Communications of the ACM. 2013 Dec 1;56(12):29-31.

Duan Y, Edwards JS, Dwivedi YK. Artificial intelligence for decision making in the era of Big Data–evolution, challenges and research agenda. International Journal of Information Management. 2019 Oct 1;48:63-71.

Dwivedi, Y. K., Hughes, L., Ismagilova, E., Aarts, G., Coombs, C., Crick, T., ... & Williams, M. D. (2019). Artificial Intelligence (AI): Multidisciplinary perspectives on emerging challenges, opportunities, and agenda for research, practice and policy. International Journal of Information Management, 101994.

Edwards, J. S., Duan, Y., & Robins, P. (2000). An analysis of expert systems for business decision making at different levels and in different roles. European Journal of Information Systems, 9(1), 36–46.

Eppinger, S., & Ulrich, K. (2015). Product design and development. McGraw-Hill Higher Education.

Fry, H. (2018). Hello world: How to be human in the age of the machine. London, UK: Transworld Publishers.

Haeffner, M., & Panuwatwanich, K. (2017). Perceived Impacts of Industry 4.0 on Manufacturing Industry and Its Workforce: Case of Germany. International conference on engineering, project, and product management (pp. 199–208). Cham: Springer (September).

Hamet P, Tremblay J. Artificial intelligence in medicine. Metabolism. 2017 Apr 1;69:S36-40.
Jha SK, Bilalovic J, Jha A, Patel N, Zhang H. Renewable energy: Present research and future scope of Artificial Intelligence. Renewable and Sustainable Energy Reviews. 2017 Sep 1;77:297-317.
Jha, S. K., Bilalovic, J., Jha, A., Patel, N., & Zhang, H. (2017). Renewable energy: Present research and future scope of Artificial Intelligence. Renewable and Sustainable Energy Reviews, 77, 297-317.
Kaplan, A. and Haenlein, M. (2019), "Siri, Siri, in my hand: who's the fairest in the land? On the interpretations, illustrations, and implications of artificial intelligence", Business Horizons, Vol. 62 No. 1, pp. 15-25.
Li, B. H., Hou, B. C., Yu, W. T., Lu, X. B., & Yang, C. W. (2017). Applications of artificial intelligence in intelligent manufacturing: A review. Frontiers of Information Technology & Electronic Engineering, 18(1), 86–96.
Mikhaylov, S. J., Esteve, M., & Campion, A. (2018). Artificial intelligence for the public sector: Opportunities and challenges of cross-sector collaboration. Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences, 376(2128).
Miller, T. (2019). Explanation in artificial intelligence: Insights from the social sciences. Artificial Intelligence, 267, 1–38.
Muhuri, P. K., Shukla, A. K., & Abraham, A. (2019). Industry 4.0: A bibliometric analysis and detailed overview. Engineering Applications of Artificial Intelligence, 78, 218–235.
Nishant R, Kennedy M, Corbett J. Artificial intelligence for sustainability: Challenges, opportunities, and a research agenda. International Journal of Information Management. 2020 Aug 1;53:102104.
Parveen, R. (2018). Artificial intelligence in construction industry: Legal issues and regulatory challenges. International Journal of Civil Engineering and Technology, 9(13), 957–962.
Pinto dos Santos D, Giese D, Brodehl S et al (2019) Medical students' attitude towards artificial intelligence: a multicentre survey. Eur Radiol 29(4):1640–1646.
PwC (2016). Blockchain - an opportunity for energy producers and consumers. PwC global power & utilities.
Recht, M. P., Dewey, M., Dreyer, K., Curtis, L., Wiro, N., Prainsack, B., & Smith, J. J. (2020). Integrating artificial intelligence into the clinical practice of radiology: challenges and recommendations. European radiology, 30(6), 3576-3584.
Rubin DL (2019) Artificial intelligence in imaging: the radiologist's role. J Am Coll Radiol 16(9):1309–1317.
Russell, S., & Norvig, P. (2021). Artificial Intelligence: A Modern Approach, Global Edition 4th. Foundations, 19, 23.
Schönberger D (2019) Artificial intelligence in healthcare: a critical analysis of the legal and ethical implications. Int J Law Info Tech 27(2):171–203
Strohm, L., Hehakaya, C., Ranschaert, E. R., Boon, W. P., & Moors, E. H. (2020). Implementation of artificial intelligence (AI) applications in radiology: hindering and facilitating factors. European radiology, 30, 5525-5532.
Tajmir SH, Alkasab TK (2018) Toward augmented radiologists: changes in radiology education in the era of machine learning and artificial intelligence. Acad Radiol 25(6):747–750.
Waltz DL. Artificial Intelligence: realizing the ultimate promises of computing. AI magazine. 1997 Sep 15;18(3):49-.
Wei, Likai. "AI-Design: Architectural Intelligent Design Approaches Based on AI." DEStech Transactions on Engineering and Technology Research icaen (2018).
Wood MJ, Teneholtz NA, Geis JR, Michalski MH, Andriole KP (2019) The need for a machine learning curriculum for radiologists. J Am Coll Radiol 16(5):740–742
Zech JR, Badgeley MA, Liu M, Costa AB, Titano JJ, Oermann (2018)Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: a cross-sectional study. PLoS Med 15(11):1002683.